

4. Mind Map Canvas

Design Phase: Architectural & Functional Mind Map

Header	Information
Date	02 October 2026
Team ID	
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Central Theme: Script-Controlled Access Control List (ACL) System

1. Requirements & User Roles

- **User Roles:**
 - bb1: Read Permission
 - bb2: Create Permission
 - bb3: Write / Edit Permission
 - bb4: Delete Permission
 - admin: System-wide overriding access
- **Target Data Structure:**
 - Table: u_institution_details
 - Key Fields: Branch (u_branch), Name, Location, Status

2. Security Architecture (ACL Layer)

- **Access Control Lists (ACLs):**
 - **Read ACL:** Evaluates dynamic script condition (u_branch == 'EEE' OR gs.hasRole('admin')).
 - **Create ACL:** Restricts record generation to users with bb2 role.
 - **Write ACL:** Restricts record modification to users with bb3 role.
 - **Delete ACL:** Restricts record deletion to users with bb4 role.

3. Execution & Evaluation Engine

- **Context Evaluation:**
 - Server-side evaluation at query execution time.
 - Row-level access check (current object inspection).

- Role validation via `gs.hasRole()`.

4. Quality Assurance & Validation

- **Testing Scenarios:**

- Impersonate User_BB1 (Branch: EEE) → Can view EEE records only.
- Impersonate User_BB1 (Branch: CSE) → Access denied / Record hidden.
- Impersonate User_BB2 → Can insert new records into `u_institution_details`.
- Impersonate User_BB3 → Can edit existing record fields.
- Impersonate User_BB4 → Can delete permitted records.
- Impersonate Admin → Full unrestricted access.